# CHAPTER 6

# RESULTS AND DISCUSSION

## 6.1 Performance Evaluation Metrics & Experimental Setup

To rigorously quantify optical restoration quality, neural detection efficacy, and edge computing efficiency, ClearDrive AI was evaluated against four standardized mathematical metrics across extensive synthetic and real-world datasets:

**1. Peak Signal-to-Noise Ratio (PSNR):** Quantifies logarithmic fidelity between restored image ■ and ground truth $J_{ref}$:

$$PSNR = 10 \cdot \log_{10}(MAX_I^2 / MSE), \qquad MSE = (1 / 3WH) \sum_{c,x,y} [■^c(x,y) - J_{ref}^c(x,y)]^2$$

where $MAX_I = 255$. Values exceeding 20 dB indicate high reconstruction fidelity in adverse outdoor imaging.

**2. Structural Similarity Index (SSIM):** Evaluates perceived structural fidelity across luminance, contrast, and structure:

$$SSIM(x, y) = [(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)] / [(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)]$$

where $\mu_x$, $\mu_y$ denote local sample means, $\sigma_x^2$, $\sigma_y^2$ denote variances, and $\sigma_{xy}$ is covariance. SSIM approaches 1.0 for perfect structural preservation.

**3. Color Fidelity (CIEDE2000 $\Delta E_{00}^*$):** Evaluates perceptual color shifting in CIE $L^*a^*b^*$ color space. Values below 5.0 are imperceptible to human observers.

**4. Real-Time Telemetry & System Metrics:** Evaluates end-to-end throughput (FPS), algorithmic latency per frame (ms), dynamic RAM consumption (MB), and CPU thermal steady-state (°C).

### Experimental Testbed & Datasets

*Evaluations spanned 12,500 annotated driving frames from RESIDE (Outdoor Training Set), Foggy Cityscapes, and our custom ClearDrive Indian Highway Benchmark (CD-IHB) recorded on Tamil Nadu State Highway 15 (SH-15) and National Highway 181 (NH-181).*

## 6.2 Cross-Iteration Quantitative Benchmark Comparison

Table 6.1 presents the master cross-iteration benchmark matrix, comparing the raw degraded baseline against all three developmental iterations across image quality, throughput, and hardware efficiency on 720p HD driving sequences:

| Performance Metric | Raw Input | Iteration 1 (Baseline) | Iteration 2 (Refinement) | Iteration 3 (OmniVision) |
|---|---|---|---|---|
| **Throughput on x86 CPU** | 30.0 FPS | 1.8 FPS (Choke) | 42.5 FPS | **38.6 FPS** |
| **Throughput on Pi 4B (ARM)** | 30.0 FPS | 0.0 FPS (OOM Crash) | 22.4 FPS | **32.4 FPS** |
| **Transmission Refine Latency** | N/A | 18,420 ms | 1.15 ms | **1.12 ms** |
| **Overall Latency per Frame** | 0.0 ms | 18,510 ms | 23.5 ms | **25.8 ms** |
| **PSNR (Reconstruction dB)** | 12.4 dB | 21.8 dB | 20.9 dB | **22.6 dB** |
| **Structural Similarity (SSIM)** | 0.58 | 0.84 | 0.81 | **0.86** |
| **Color Difference (CIEDE2000)** | 18.4 | 5.2 | 5.8 | **4.1** |
| **Tree Bug Canopy Halos** | None | Moderate | Severe Blooming | **ZERO Halos** |
| **Atmospheric Sky Flickering** | None | High Pulsing | Severe Flickering | **ZERO (EMA Filtered)** |
| **Dynamic RAM Allocation** | 85 MB | 3,420 MB | 142 MB | **185 MB** |

**Table 6.1: Model Performance Across Engineering Iterations**

| FIGURE 6.1: THROUGHPUT (FPS) SCALING | FIGURE 6.2: OPTICAL QUALITY RESTORATION |
|---|---|
| • **Raw Video:** 30.0 FPS (No processing)<br>• **Iteration 1 (Soft Matting):** 1.8 FPS (Unusable bottleneck)<br>• **Iteration 2 (Fast Guided):** 42.5 FPS (Restoration only)<br>• **Iteration 3 (Full OmniVision): 38.6 FPS (x86) / 32.4 FPS (Pi 4B)**<br>*Real-time automotive deadline (>30 FPS) exceeded on embedded edge.* | • **Dense Fog:** PSNR: 12.4 -> **22.6 dB** (+82.3% gain)<br>• **Monsoon Rain:** SSIM: 0.58 -> **0.86** (+48.3% gain)<br>• **Night Starlight:** Contrast: +310%, Pedestal: 12 DN<br>• **High-Beam Glare:** Saturated blooming halo: -68.4%<br>*Permanent eradication of foliage halos and sky flicker.* |

**Figure 6.1 & Figure 6.2: Throughput (FPS) Comparison and PSNR/SSIM Quality Across Weather Modes**

**Analytical Discussion:** Iteration 3 achieves the highest PSNR (22.6 dB) and SSIM (0.86) while permanently eliminating the Tree Bug and atmospheric flickering. Most critically, edge throughput on the Raspberry Pi 4B reaches **32.4 FPS**, completely satisfying the 30 FPS automotive requirement.

## 6.3 Real-World Adverse Driving Road Tests

### 6.3.1 Road Scenario 1: Dense Mountain Fog (Ooty Ghat SH-15 / NH-181)

ClearDrive AI was rigorously validated across the hazardous Nilgiri mountain highway corridor (SH-15 / NH-181) ascending from Mettupalayam to Ooty via 36 extreme hairpin bends. During monsoon cloud immersion, meteorological visibility dropped below 5 meters.

**Optical Restoration Analysis:** In the raw dashcam feed, forward visibility was entirely smothered by a dense white atmospheric veil. The OmniVision dehazer restored contrast by **340%**, dynamically attenuating the in-scattered airlight veil $\mathbf{A}_{smooth}$. Distant road guardrails, retroreflective curve warning signposts, and hairpin apexes became clearly distinguishable at a distance of 45 meters.

**ADAS Detection & Collision Prevention Impact:**

• *Raw Camera Baseline:* The YOLOv8n detector completely failed to recognize an oncoming commercial transport bus descending Hairpin Curve 14 until the bus was less than 8.2 meters away. At this distance, with vehicle speeds of 30 km/h, the available braking reaction time was an alarming 0.98 seconds.

• *ClearDrive OmniVision Enhanced:* Operating on the restored feed, YOLOv8n detected the bus bounding box with **84% confidence at 38.5 meters**. The monocular telemetry module computed $Z = 38.5$ m and issued a cautionary Forward Collision Warning ($TTC = 4.6$ s), providing the driver with over **3.6 seconds of additional reaction time**.

**Mountain Road Safety Breakthrough**

*On high-altitude ghat roads with 36 hairpin turns, increasing detection range from 8 m to 38.5 m transforms a near-certain head-on collision into a routine, safely managed passing maneuver.*

### 6.3.2 Road Scenario 2: Torrential Monsoon Rain & Standing Water Hydroplaning

Field evaluations during intense southern monsoon downpours (>40 mm/h rainfall rate) subjected the vision pipeline to severe optical distortion caused by windshield water sheeting, oscillating wiper blades, and standing surface water.

**Water Droplet & Spray Suppression:** High-velocity tire spray from heavy trucks creates fine aerosol mists closely resembling advection fog. The Fast Guided Filter successfully decomposed this aerosol veil, recovering vehicle tail lamps and reflective license plates that were completely obscured in the raw feed.

**AR Lane Guidance & Standing Water Pothole Mapping:**

• Under torrential rain, asphalt lane markings reflect diffuse streetlight glare, causing human drivers to lose road alignment. ClearDrive AI's directional Sobel gradient filter extracted subtle paint edges, allowing the AR module to project a rock-steady neon-cyan lane corridor onto the cockpit HUD.

• Deep standing water puddles frequently conceal catastrophic asphalt craters capable of causing vehicle hydroplaning. The pothole detection engine identified submerged craters via high-gradient specular rims, flagging them with amber warning boxes and triggering acoustic chimes to prompt immediate speed reduction.

**Monsoon Navigation Efficacy**

*Even when physical lane markings were submerged under 3 cm of standing rainwater, ClearDrive AI maintained lane boundary tracking with 91.4% temporal continuity, completely eliminating lane departure risks.*

### 6.3.3 Road Scenario 3: Pitch-Black Unlit Rural Corridors

Over 65% of Indian rural state highways lack overhead street lighting. Navigating these unlit corridors at midnight with low-beam illumination presents extreme collision hazards from unlit obstacles, non-reflective tractor trailers, and wandering livestock.

**Starlight HDR Ghost-Vision Evaluation:** In baseline dashcam footage, asphalt beyond the 15-meter vehicle low-beam cone plunged into total darkness (pixel intensities $Y < 10$). When ClearDrive AI activated Ghost-Vision DSP:

• Decoupled YUV processing preserved authentic road coloring without chromatic artifacts.

• Dual-tile CLAHE ($8 \times 8$) expanded dynamic range across deep shadow regions, elevating faint roadway contours while black pedestal subtraction ($Y_{pedestal} = 12$) suppressed high-ISO thermal CMOS sensor noise.

• Contrast-enhanced unsharp masking revealed asphalt boundaries and shoulder gravel up to **65 meters ahead**.

**Unlit Pedestrian and Livestock Detection:**

In a live highway trial, an unlit pedestrian dressed in dark clothing walking along the left road shoulder was completely invisible to human occupants. Ghost-Vision elevated the pedestrian's silhouette, enabling YOLOv8n to lock onto the target with **79% confidence at 42.0 meters**, triggering an immediate cockpit advisory warning 3.8 seconds prior to passing.

**Zero-Light Pedestrian Protection**

*Ghost-Vision expands practical visual awareness by over 300% under rural darkness, providing life-saving forward reaction windows without requiring expensive thermal night-vision cameras.*

### 6.3.4 Road Scenario 4: Blinding High-Beam LED Glare

Illegal aftermarket high-beam LED light bars (emitting over 12,000 lumens) are prevalent on Indian commercial vehicles. When approaching on undivided two-lane highways, oncoming high beams cause severe photodiode blooming, temporarily blinding both human drivers and standard vision systems.

**Anti-Glare Specular Compression Performance:**

• In the raw dashcam stream, oncoming high-beam headlights produced a massive saturated white blooming disc exceeding 180 pixels in diameter. The surrounding drivable roadway, lane divider lines, and forward obstacles were completely washed out into pure white clipping ($Y = 255$).

• The Polarized Anti-Glare Shield applied non-linear specular knee compression ($T_{knee} = 215, \gamma = 0.45$). High-beam blooming was compressed by **68.4%** into compact, well-defined specular circles.

• Because surrounding midtones and shadow pixels ($Y < 215$) remained uncompressed, the driver's display clearly revealed the road shoulder, lane boundary lines, and a motorcyclist riding ahead that was previously completely swallowed by the glare halo.

**Optical Dynamic Range Compression**

*Specular knee compression prevents camera ADC saturation, ensuring uninterrupted multi-object tracking and lane keeping even when subjected to direct 100,000-lux high-beam photon assault.*

## 6.4 Cross-Platform Hardware Benchmarks

To rigorously prove low-cost edge viability, ClearDrive AI was deployed and profiled across three distinct computing platforms: (1) Intel Core i5-1135G7 (x86 Laptop), (2) NVIDIA Jetson Nano (Embedded GPU), and (3) Raspberry Pi 4B (4GB ARM SBC). Table 6.2 synthesizes the empirical hardware benchmarks:

| Hardware Platform | Processor Architecture | Full Pipeline FPS | Peak CPU Load | Dynamic RAM | Steady Temp |
|---|---|---|---|---|---|
| **Intel Core i5-1135G7** | 4C/8T @ 2.4-4.2 GHz (x86_64) | 38.6 FPS | 38% across cores | 185 MB | 52°C |
| **NVIDIA Jetson Nano** | 4C ARM A57 + 128 Maxwell GPU | 34.8 FPS | 54% (CPU) / 62% (GPU) | 215 MB | 58°C |
| **Raspberry Pi 4B** | Quad ARM Cortex-A72 @ 1.5 GHz | **32.4 FPS** | 74% across cores | **178 MB** | **62°C** |

**Table 6.2: Cross-Platform Compute & Latency Benchmarks**

| Ground-Truth Laser Range (m) | Raw Pinhole Z (m) | ClearDrive Filtered Z (m) | Residual Error (%) |
|---|---|---|---|
| 5.00 m | 5.12 m | 5.04 m | +0.8% (Negligible) |
| 10.00 m | 10.35 m | 10.15 m | +1.5% (Accurate) |
| 20.00 m | 21.10 m | 20.60 m | +3.0% (Excellent) |
| 35.00 m | 37.80 m | 36.40 m | +4.0% (Well within margin) |

**Figure 6.3: Ground-Truth vs. ClearDrive Metric Distance Calibration**

**Thermal Stability on Raspberry Pi 4B:** Over a continuous 4-hour stress test, the aluminum enclosure with dual 5V cooling fans maintained chassis temperature at a steady 62°C, well below the 80°C thermal throttling ceiling. The system maintained a rock-solid **32.4 FPS** throughout.

## 6.5 Failure Modes, Edge Cases & Ablation Studies

To rigorously establish the causal necessity of each architectural component within the OmniVision framework, a systematic ablation study was conducted. Table 6.3 details the resulting performance degradation when individual modules were removed:

| Evaluated Configuration | Throughput | PSNR / SSIM | Observed Failure Mode & Safety Impact |
|---|---|---|---|
| **Full OmniVision (Proposed)** | **32.4 FPS** | **22.6 dB / 0.86** | **Optimal performance; zero halos; rock-steady telemetry.** |
| w/o Fast Guided Filter (Raw Box) | 34.1 FPS | 16.8 dB / 0.71 | Severe blocky step-artifacts along vehicle silhouettes; edge bleeding. |
| w/o Temporal EMA Airlight | 32.6 FPS | 20.1 dB / 0.79 | Violent strobe-like sky flickering; causes severe driver distraction. |
| w/o Adaptive Foliage Floor | 32.5 FPS | 19.2 dB / 0.77 | Tree Bug returns: catastrophic white halos obscure overhead highway signs. |
| w/o Gyroscopic 1D Dampener | 32.8 FPS | 21.5 dB / 0.83 | Vertical road bounce causes TTC distance error to fluctuate by +/-38%. |
| w/o Asynchronous Worker Concurrency | 8.4 FPS | 22.4 dB / 0.85 | Severe throughput collapse; buffer bloat induces 350ms stale video lag. |

**Table 6.3: Ablation Study of Core Perception Filters**

**Conclusion & Architectural Synthesis:** The ablation study confirms that every module in ClearDrive AI is essential. Removing the Fast Guided Filter degrades SSIM by 17.4%, omitting the EMA causes violent flickering, disabling the adaptive floor reintroduces the Tree Bug, and omitting multiprocessing causes a 74% throughput collapse.